Chapter 1

Language of category theory

1.1 Universe

1.1.1 Definition. We call *universe* any set $\mathcal{U}$ whose elements are sets, which satisfies the following conditions:

(a) $\varnothing \in \mathcal{U}$;

(b) If $A \in \mathcal{U}$ and $B \in A$, then $B \in \mathcal{U}$ (in particular, B is also a set);

(c) If $A \in \mathcal{U}$, then $\mathscr{P}(A) \in \mathcal{U}$;

(d) If $I \in \mathcal{U}$ and $(A_i)_{i \in I} \in \mathcal{U}^I$, then $\bigcup_{i \in I} A_i \in \mathcal{U}$.

1.1.2 Axioms. Every set belongs to some universe.

1.1.3 Proposition. *Let $\mathcal{U}$ be a universe.*

(1) *If $A \in \mathcal{U}$ and $B \subseteq A$, then $B \in \mathcal{U}$.*

(2) *For any $A \in \mathcal{U}$, one has $\{A\} \in \mathcal{U}$.*

(3) *If A and B are elements of $\mathcal{U}$, then $\{A, B\} \in \mathcal{U}$.*

(4) *If A and B are elements of $\mathcal{U}$, then the ordered pair (A, B), that is defined as*
$$\{\{A\}, \{A, B\}\},$$
belongs to $\mathcal{U}$.

(5) *If X and Y are elements of $\mathcal{U}$, then the Cartesian product $X \times Y$, that is defined as*
$$\{(x, y) \mid x \in X,\ y \in Y\},$$
belongs to $\mathcal{U}$.

(6) *If X and Y are elements of $\mathcal{U}$, then the set of all correspondences form X to Y belongs to $\mathcal{U}$.*

(7) *If X and Y are elements of U, then $Y^X \in \mathcal{U}$.*

(8) *Let I be a non-empty set in $\mathcal{U}$ and let $(X_i)_{i\in I} \in \mathcal{U}^I$, then $\prod_{i\in I} X_i$ belongs to $\mathcal{U}$.*

Proof. (1) By the condition (c) in Definition 1.1.1, one has $\mathscr{P}(A) \in \mathcal{U}$. Since B is an element of $\mathscr{P}(A)$, one has $B \in \mathcal{U}$.

(2) By the condition (c) in Definition 1.1.1, one has $\mathscr{P}(A) \in \mathcal{U}$. Since $\{A\}$ is a subset of $\mathscr{P}(A)$, by (1) we deduce that $\{A\} \in \mathcal{U}$.

(3) By the condition (c) in Definition 1.1.1, one has

$$\mathscr{P}(\mathscr{P}(\varnothing)) = \{\varnothing, \{\varnothing\}\} \in \mathcal{U}.$$

By (2), one has $\{A\} \in \mathcal{U}$ and $\{B\} \in \mathcal{U}$. Hence, by the condition (d) in Definition 1.1.1, we deduce

$$\{A, B\} = \{A\} \cup \{B\} \in \mathcal{U}.$$

(4) By (2), one has $\{A\} \in \mathcal{U}$. By (3), one has $\{A, B\} \in \mathcal{U}$. Still by (3), we deduce that

$$(A, B) := \{\{A\}, \{A, B\}\} \in \mathcal{U}.$$

(5) For any $x \in X$ and $y \in Y$, by the condition (b) of Definition 1.1.1, one has $x \in \mathcal{U}$ and $y \in \mathcal{U}$. By (4) one has $(x, y) \in \mathcal{U}$. Hence, by (2) one has $\{(x, y)\} \in \mathcal{U}$. For any $x \in X$, by (4) one has

$$\{x\} \times Y = \bigcup_{y\in Y} \{(x, y)\} \in \mathcal{U}.$$

Therefore, by the condition (d) in Definition 1.1.1, we deduce

$$X \times Y = \bigcup_{x\in X} \{x\} \times Y \in \mathcal{U}.$$

(6) Note that a correspondence from X to Y is of the form $f = ((X, Y), \Gamma)$, where Γ is an element of $\mathscr{P}(X \times Y)$. By (5), $X \times Y \in \mathcal{U}$. By (1), we deduce that $\Gamma \in \mathcal{U}$. Hence, by (4) we obtain that $((X, Y), \Gamma) \in \mathcal{U}$ and then by (2) we get

$$\{((X, Y), \Gamma)\} \in \mathcal{U}.$$

By the condition (c) in Definition 1.1.1 and the fact $X \times Y \in \mathcal{U}$, we deduce that $\mathscr{P}(X \times Y) \in \mathcal{U}$. Therefore, by the condition (d) in Definition 1.1.1, the set of all correspondences from X to Y, which can be written as

$$\bigcup_{\Gamma\in\mathscr{P}(X\times Y)} \{((X, Y), \Gamma)\},$$

belongs to $\mathcal{U}$.

(7) follows from (6) and (1) since Y^X is a subset of the set of all correspondences from X to Y.

(8) Let $X = \bigcup_{i\in I} X_i$. Note that $\prod_{i\in I} X_i$ is a subset of X^I. By (7), X^I belongs to $\mathcal{U}$. Hence, by (1), $\prod_{i\in I} X_i \in \mathcal{U}$. □

1.2 Category

1.2.1 Definition. Let $\mathcal{U}$ be a universe. We call $\mathcal{U}$-*category* $\mathcal{C}$ the data that consist of:

(i) a set $\mathrm{Obj}(\mathcal{C})$ (the elements of which are called *objects*);

(ii) for any $(X, Y) \in \mathrm{Obj}(\mathcal{C}) \times \mathrm{Obj}(\mathcal{C})$, a set $\mathcal{C}(X, Y) \in \mathcal{U}$ (the elements of which are called *morphisms* from X to Y);

(iii) for any triple $(X, Y, Z) \in \mathrm{Obj}(\mathcal{C})^3$, a composition mapping

$$\mathcal{C}(X, Y) \times \mathcal{C}(Y, Z) \longrightarrow \mathcal{C}(X, Z), \quad (f, g) \longmapsto g \circ f,$$

satisfying the following axioms:

(a) for any $X \in \mathrm{Obj}(\mathcal{C})$, there exists a morphism $\mathrm{Id}_X \in \mathcal{C}(X, X)$ (called the *identity morphism* of X) such that, for any object Y of $\mathcal{C}$, one has

$$\begin{aligned} \forall\, f \in \mathcal{C}(X, Y), &\quad f \circ \mathrm{Id}_X = f, \\ \forall\, g \in \mathcal{C}(Y, X), &\quad \mathrm{Id}_X \circ g = g, \end{aligned}$$

that, is, $f \circ \mathrm{Id}_X = f$ and $\mathrm{Id}_X \circ g = g$ whenever defined.

(b) for any $(X, Y, Z, W) \in \mathrm{Obj}(\mathcal{C})^4$, $f \in \mathcal{C}(X, Y)$, $g \in \mathcal{C}(Y, Z)$, $h \in \mathcal{C}(Z, W)$, one has

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

If $\mathrm{Obj}(C)$ belongs to $\mathcal{U}$, then we say that $\mathcal{C}$ is $\mathcal{U}$-*small*.

If $\mathrm{Obj}(C)$ is finite and if $\mathcal{C}(X, Y)$ is finite for any $(X, Y) \in \mathrm{Obj}(C)^2$, then we say that $\mathcal{C}$ is a *finite category*.

1.2.2 Example. Let $\mathcal{U}$ be a universe. Then the sets in $\mathcal{U}$ and the mappings between sets form a category $\mathcal{U}$-**Set**. The set of objects of the category $\mathcal{U}$-**Set** is $\mathcal{U}$. For any $(A, B) \in \mathcal{U} \times \mathcal{U}$, the set

$$\mathcal{U}\text{-}\mathbf{Set}(A, B)$$

is just the set of all mappings from A to B.

1.2.3 Remark. In practice we often omit the mention of the universe $\mathcal{U}$ in which we consider the category theory. With this abbreviation, a $\mathcal{U}$-category is simply called a *category*. A $\mathcal{U}$-small $\mathcal{U}$-category is simply called a *small category*. Similarly, the category of sets in $\mathcal{U}$ and the mappings between sets is denoted as **Set**.

1.2.4 Notation. (1) We denote by **Top** the category of topological spaces and continuous mappings between topological spaces.

(2) We denote by **Gp** the category of groups and homomorphisms of groups. We dnote by **Mon** the cateogry of monoids and homomorphisms of monoids. We denote by **Ab** the category of abelian groups and homomorphism of groups.

(3) We denote by **Ring** the category of unitary rings and homomorphism of unitary rings, and by **CRing** the category of commutative, unitary rings and homomorphisms of unitary rings.

(4) Let K be a unitary ring. We denote by ${}_K\mathbf{Mod}$ the category of left K-modules. If M and N are left K-modules, we usually denote the set of left K-module homomorphisms from M to N as $\mathrm{Hom}_K(M, N)$.

(5) Let K be a commutative unitary ring. We denote by $\mathbf{Alg}_K$ the category of K-algebras and homomorphism of K-algebras. Similarly we denote by $\mathbf{CAlg}_K$ the category of commutative K-algebras and homomorphism of K-algebras.

1.2.5 Remark. Let K be a commutative unitary ring. Recall that a K-algebra is by definition a K-module A together with a bilinear mapping

$$A \times A \longrightarrow A$$

such that A equipped with this bilinear mapping forms a unitary ring.

1.2.6 Notation. Let $\mathcal{C}$ be a category, X and Y be objects of $\mathcal{C}$. We often use the expression

$$f : X \longrightarrow Y$$

or

$$X \xrightarrow{f} Y$$

to denote a morphism f in $\mathcal{C}(X, Y)$.

1.2.7 Proposition. *Let $\mathcal{C}$ be a category and X be an object of $\mathcal{C}$. The identity morphism of X is unique.*

Proof. Let Id_X and Id'_X be both identity morphisms of X. By definition one has

$$\mathrm{Id}_X = \mathrm{Id}_X \circ \mathrm{Id}'_X = \mathrm{Id}'_X$$

□

1.2.8 Definition. Let $\mathcal{C}$ be a category and $(X, Y) \in \mathrm{Obj}(\mathcal{C})^2$. Let $f \in \mathcal{C}(X, Y)$. We say that f is *left invertible* if there exists $g : Y \to X$ such that

$$g \circ f = \mathrm{Id}_X,$$

In this case we say that g is a *left inverse* of f. We say that f is *right invertible* if there exists $h : Y \to X$ such that

$$f \circ h = \mathrm{Id}_Y,$$

In this case we say that h is a *right inverse* of f.

1.2.9 Proposition. *Let $\mathcal{C}$ be a category and $(X, Y) \in \mathrm{Obj}(\mathcal{C})^2$. Let $f \in \mathcal{C}(X, Y)$. Suppose that f is left invertible and right invertible. Then f has only one left inverse and one right inverse, and the left inverse and right inverse of f are equal to each other.*

Proof. Let g be a left inverse and h be a right inverse of f. One has

$$g = g \circ \mathrm{Id}_Y = g \circ (f \circ h) = (g \circ f) \circ h = \mathrm{Id}_X \circ h = h$$

□

1.2.10 Definition. Let $\mathcal{C}$ be a category and $(X, Y) \in \mathrm{Obj}(\mathcal{C})^2$. Let $f \in \mathcal{C}(X, Y)$. If f is left invertible and right invertible, then there exists a unique morphism $f^{-1} \in \mathcal{C}(Y, X)$ such that

$$f \circ f^{-1} = \mathrm{Id}_Y, \quad f^{-1} \circ f = \mathrm{Id}_X\,.$$

In this case we say that f is an *isomorphism* and that f^{-1} is the *inverse* of f.

If there exists an isomorphism in $\mathcal{C}(X, Y)$, then we say that X and Y are *isomorphic*.

1.2.11 Example. We consider a category $\mathcal{C}$ having only one object $*$, namely $\mathrm{Obj}(\mathcal{C})$ is the one point set $\{*\}$. In this case, $\mathcal{C}(*, *)$ forms a monoid under the composition law. The identity morphism of $*$ is the neutral element of this monoid, and isomorphisms are just invertible element of the monoid.

1.2.12 Proposition. *Let $\mathcal{C}$ be a category. If $f : X \to Y$ and $g : Y \to Z$ are isomorphisms, then $g \circ f$ is an isomorphism, and*

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Proof. One has

$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \mathrm{Id}_Y \circ f = f^{-1} \circ f = \mathrm{Id}_X,$$
$$(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ \mathrm{Id}_Y \circ g^{-1} = g \circ g^{-1} = \mathrm{Id}_Z\,.$$

□

1.2.13 Definition. Let $\mathcal{C}$ be a category. We define the *opposite category* $\mathcal{C}^{\mathrm{op}}$ of $\mathcal{C}$ as follows: the category $\mathcal{C}^{\mathrm{op}}$ has the same objects as $\mathcal{C}$, and, for any $(X, Y) \in \mathrm{Obj}(\mathcal{C}) \times \mathrm{Obj}(\mathcal{C})$,

$$\mathcal{C}^{\mathrm{op}}(X, Y) := \mathcal{C}(Y, X).$$

The composition mapping

$$\mathcal{C}^{\mathrm{op}}(X, Y) \times \mathcal{C}^{\mathrm{op}}(Y, Z) \longrightarrow \mathcal{C}^{\mathrm{op}}(X, Z)$$

is given by the composition mapping

$$\mathcal{C}(Y, X) \times \mathcal{C}(Z, Y) \longrightarrow \mathcal{C}(Z, X), \quad (f, g) \longmapsto f \circ g$$

in the structure of $\mathcal{C}$.

Note that, if we use an arrow $X \xrightarrow{f} Y$ to denote a morphism in $\mathcal{C}$, it suffices to reverse the arrow to denote the same morphism viewed as a morphism in $\mathcal{C}^{\mathrm{op}}$.

1.2.14 Definition. A diagram of morphisms in a category is a directed graph whose vertices are objects and arrows are morphisms. A diagram is *commutative* if for any two vertices X, Y, the morphisms obtained by composing morphisms along different paths coincide. For example, in a category $\mathcal{C}$, the diagram

$$\begin{tikzcd} & Y \arrow[dr, "g"] & \\ X \arrow[ur, "f"] \arrow[rr, "h"'] & & Z \end{tikzcd}$$

is commutative if and only $h = g \circ f$. The diagram

$$\begin{tikzcd} X \arrow[r, "a"] \arrow[d, "b"'] & Y \arrow[d, "c"] \\ Z \arrow[r, "d"'] & W \end{tikzcd}$$

is commutative if and only if

$$d \circ b = c \circ a.$$

1.3 Functors

1.3.1 Definition. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. We call *functor* from $\mathcal{C}$ to $\mathcal{D}$ any data that consist of:

(i) a mapping on objects $F : \mathrm{Obj}(\mathcal{C}) \to \mathrm{Obj}(\mathcal{D})$;

(ii) for any pair $(X, Y) \in \mathcal{C} \times \mathcal{C}$, a mapping of sets

$$F : \mathcal{C}(X, Y) \to \mathcal{D}(FX, FY),$$

such that:

(a) $F(\mathrm{Id}_X) = \mathrm{Id}_{F(X)}$;

(b) for any morphisms

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

in $\mathcal{C}$, $F(g \circ f) = F(g) \circ F(f)$.

Usually, we use the expression $F : \mathcal{C} \to \mathcal{D}$ to denote a functor F from $\mathcal{C}$ to $\mathcal{D}$.

1.3.2 Example. Consider

$$P : \mathbf{CRing} \to \mathbf{Set}$$

such that for all all $A \in \mathrm{Obj}(\mathbf{CRing})$

$$P(A) = \{(x, y, z) \in A^3 \,|\, x^2 + y^2 = z^2\}$$

and for all homomorphism $f : A \to B$ of commutative rings,

$$P(f) : P(A) \longrightarrow P(B), \quad (x, y, z) \longmapsto (f(x), f(y), f(z))$$

1.3.3 Example. Let $\mathcal{C}$ be a category and X be an object of $\mathcal{C}$. We denote by h_X the functor from $\mathcal{C}$ to **Set** that sends any object Y of $\mathcal{C}$ to the set $\mathcal{C}(X, Y)$, and send any morphism $f : Y \to Z$ to the mapping

$$h_X(f) : \mathcal{C}(X, Y) \longrightarrow \mathcal{C}(X, Z), \quad g \longmapsto f \circ g.$$

1.3.4 Example. Let $\mathcal{C}$ be a category, and $\mathrm{Id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ be a mapping that send $X \in \mathrm{Obj}(\mathcal{C})$ to X and $f : X \to Y$ to f. This is a functor, called the *identity functor* of $\mathcal{C}$.

1.3.5 Example. Let $\mathcal{C}, \mathcal{D}, \mathcal{E}$ be categories, $F : \mathcal{C} \to \mathcal{D}$ be functors. We define a functor

$$G \circ F : \mathcal{C} \longrightarrow \mathcal{E}$$

such that for all $X \in \mathrm{Obj}(\mathcal{C})$,

$$(G \circ F)(X) = G(F(X))$$

and for all morphism $f : X \to Y$ in $\mathcal{C}$,

$$(G \circ F)(f) = G(F(f)).$$

1.3.6 Definition. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, F and G be functors from $\mathcal{C}$ to $\mathcal{D}$. We call *natural transformation* (morphism of functors)from F to G any family

$$(\varphi_X : F(X) \longrightarrow G(X))_{X \in \mathrm{Obj}(\mathcal{C})}$$

such that, for any morphism $f : X \to Y$ in $\mathcal{C}$, the diagram

$$\begin{tikzcd} F(X) \arrow[r, "\varphi_X"] \arrow[d, "F(f)"'] & G(X) \arrow[d, "G(f)"] \\ F(Y) \arrow[r, "\varphi_Y"'] & G(Y) \end{tikzcd}$$

is commutative, namely

$$G(f) \circ \varphi_X = \varphi_Y \circ F(f).$$

We denote by $\mathrm{Nat}(F, G)$ the set of all natural transformations from F to G. We often use the expression $\varphi : F \to G$ to signify that φ is a natural transformation from F to G.

If F, G and H are functors from $\mathcal{C}$ to $\mathcal{D}$, $\varphi = (\varphi_X)_{X \in \mathrm{Obj}(\mathcal{C})}$ is a natural transformation from F to G, and $\psi = (\psi_X)_{X \in \mathrm{Obj}(\mathcal{C})}$ is a natural transformation from G to H, then

$$\psi \circ \varphi := (\psi_X \circ \varphi_X)_{X \in \mathrm{Obj}(\mathcal{C})}$$

is a natural transformation from F to H. We call it the *composite* of ψ and φ.

1.3.7 Example. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, F be a functor from $\mathcal{C}$ to $\mathcal{G}$. Then $(\mathrm{Id}_{F(X)})_{X \in \mathrm{Obj}(\mathcal{C})}$ is a natural transformation from F to itself. It is denoted as Id_F and is called the *identity natural transformation* of F.

1.3.8 Remark. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Let $\mathbf{Fun}(\mathcal{C}, \mathcal{D})$ be the data composed of the set (which does not necessarily belong to the universe with respect to which the cateogires $\mathcal{C}$ and $\mathcal{D}$ are considered) of all functors from $\mathcal{C}$ to $\mathcal{D}$ and natural transformations betwen them. Then $\mathbf{Fun}(\mathcal{C}, \mathcal{D})$ forms a category (with respect to some suitable universe). It is called the category of functors from $\mathcal{C}$ to $\mathcal{D}$.

1.3.9 Definition. Let $\mathcal{C}$ and $\mathcal{D}$ be two categories, F and G be functors from $\mathcal{C}$ to $\mathcal{D}$. If every φ_X, $X \in \mathrm{Obj}(\mathcal{C})$ is an isomorphism, we say that φ is a *natural isomorphism*. In this case the family

$$(\varphi_X^{-1} : G(X) \to F(X))_{X \in \mathrm{Obj}(\mathcal{C})}$$

is a natural transformation from G to F. It is denoted as φ^{-1} and is called the inverse of φ. Note that

$$\varphi^{-1} \circ \varphi = \mathrm{Id}_F, \quad \varphi \circ \varphi^{-1} = \mathrm{Id}_G\,.$$

If there exists a natural isomorphism from F to G, we say that the functors F and G are *isomorphic*.

1.3.10 Remark. An isomorphism of functors between the categories $\mathcal{C}$ and $\mathcal{D}$ is an isomorphism in the category $\mathbf{Fun}(\mathcal{C}, \mathcal{D})$.

1.3.11 Proposition. *Let $\mathcal{U}$ be a universe, $\mathcal{C}$ be a small $\mathcal{U}$-category and $\mathcal{D}$ be a $\mathcal{U}$-category. Then for any functors F and G from $\mathcal{C}$ to $\mathcal{D}$, the set of natural transformations from F to G belongs to $\mathcal{U}$.*

Proof. The set of natural transformations from F and G is a subset of

$$\prod_{X \in \mathrm{obj}(\mathcal{C})} \mathcal{D}(F(X), G(X)).$$

Since $\mathcal{C}$ is assumed to be a small category, one has $\mathrm{Obj}(\mathcal{C}) \in \mathcal{U}$. Moreover, since $\mathcal{D}$ is a $\mathcal{U}$-category, for any $X \in \mathrm{Obj}(\mathcal{C})$, one has

$$\mathcal{D}(F(X), G(X)) \in \mathcal{U}.$$

Hence

$$\prod_{X \in \mathrm{Obj}(\mathcal{C})} \mathcal{D}(F(X), G(X)) \in \mathcal{U}.$$

Therefore, by (1) of Proposition 1.1.3, the set of natural transformations from F to G belongs to $\mathcal{U}$.

□

1.3.12 Corollary. *Let $\mathcal{U}$ be a universe. Let $\mathcal{C}$ be a $\mathcal{U}$-small category and $\mathcal{D}$ be a $\mathcal{U}$-category. Then $\mathbf{Fun}(\mathcal{C}, \mathcal{D})$ is a $\mathcal{U}$-category.*

1.3.13 Remark. In the particular case where $\mathcal{D}$ is the category of sets, the functor category $\mathbf{Fun}(\mathcal{C}, \mathbf{Set})$ is also denoted as $\widehat{\mathcal{C}}$.

1.4 Yoneda Lemma

1.4.1 Theorem (Yoneda Lemma)**.** *Let $\mathcal{C}$ be a category, F be a functor from $\mathcal{C}$ to* **Set***, and $X \in \mathcal{C}$. Then the mapping*

$$\beta : \mathrm{Nat}(h_X, F) \longrightarrow F(X), \qquad \psi \longmapsto \psi_X(\mathrm{Id}_X)$$

is a bijection.

Proof. We construct a mapping

$$\alpha : F(X) \longrightarrow \mathrm{Nat}(h_X, F)$$

as follows. Given $x \in F(X)$, we define $\alpha(x)$ as the family

$$\alpha(x)_Y : \mathcal{C}(X, Y) \to F(Y), \qquad f \longmapsto F(f)(x),$$

where $Y \in \mathrm{Obj}(\mathcal{C})$.

We check that α is a natural transformation, namely, for any morphism $u : A \to B$ in $\mathcal{C}$, one has

$$F(u) \circ \alpha(x)_A = \alpha(x)_B \circ h_X(u).$$

Indeed, for any $f \in \mathcal{C}(X, A)$,

$$\begin{aligned}(F(u) \circ \alpha(x)_A)(f) = F(u)\big(F(f)(x)\big) &= F(u \circ f)(x) \\ &= \alpha(x)_B(u \circ f) = (\alpha(x)_B \circ h_X(u))(f).\end{aligned}$$

Now we show these two mappings α and β are inverse to each other. Let $\psi \in \mathrm{Nat}(h_X, F)$ and $x = \psi_X(\mathrm{Id}_X)$. Since the diagram

$$\begin{tikzcd} \mathcal{C}(X,X) \arrow[r, "\psi_X"] \arrow[d, "h_X(f)"'] & F(X) \arrow[d, "F(f)"] \\ \mathcal{C}(X,Y) \arrow[r, "\psi_Y"'] & F(Y) \end{tikzcd}$$

is commutative, for any $f : X \to Y$, one has

$$\alpha(x)_Y(f) = F(f)(x) = F(f)\big(\psi_X(\mathrm{Id}_X)\big) = \psi_Y(h_X(f)(\mathrm{Id}_X)) = \psi_Y(f).$$

Hence $\alpha(x) = \psi$. Conversely,

$$\alpha(x)_X(\mathrm{Id}_X) = F(\mathrm{Id}_X)(x) = \mathrm{Id}_{F(X)}(x) = x.$$

□

1.4.2 Notation. Let $\mathcal{C}$ be a category, X and Y be objects of $\mathcal{C}$ and $f : X \to Y$ be a morphism. We denote by $\upsilon(f)$ the natural transfrom from h_Y to h_X corresponding to f by the bijection

$$\mathrm{Nat}(h_Y, h_X) \longrightarrow h_X(Y) = \mathcal{C}(X, Y)$$

defined by Yoneda lemma.

Note that, for any $W \in \mathrm{obj}\,\mathcal{C}$,

$$\upsilon(f)_W : \mathcal{C}(Y, W) \longrightarrow \mathcal{C}(X, W)$$

sends $u \in \mathcal{C}(Y, W)$ to $u \circ f$ (see the construction of the mapping α in the proof of Theorem 1.4.1).

1.4.3 Proposition. *Let $\mathcal{C}$ be a category.*

(1) *Let X be an object of $\mathcal{C}$. One has*

$$\mathrm{Id}_{h_X} = \upsilon(\mathrm{Id}_X).$$

(2) *Let X, Y and Z be objects of $\mathcal{C}$, $f : X \to Y$ and $g : Y \to Z$ be morphisms. Then*

$$\upsilon(g \circ f) = \upsilon(f) \circ \upsilon(g).$$

(3) *Let X and Y be object of $\mathcal{C}$, and $f : X \to Y$ be a morphism. Then f is an isomorphism if and only if $\upsilon(f)$ is an isomorphism. Moreover, in the case where f is an isomorphism, one has*

$$\upsilon(f^{-1}) = \upsilon(f)^{-1}.$$

Proof. (1) For any $W \in \mathrm{obj}\,\mathcal{C}$ and any $u \in \mathcal{C}(X, W)$, one has

$$\upsilon(\mathrm{Id_X})_W(u) = u \circ \mathrm{Id}_X = u.$$

Hence $\upsilon(\mathrm{Id_X}) = \mathrm{Id}_{h_X}$.

(2) For any $W \in \mathrm{obj}\,\mathcal{C}$ and any $u \in \mathcal{C}(Z, W)$, one has

$$\upsilon(g \circ f)_W(u) = u \circ (g \circ f) = (u \circ g) \circ f = \upsilon(f)_W(u \circ g) = \upsilon(f)_W(\upsilon(g)_W(u)).$$

(3) Suppose that f is an isomorphism. Then, by (2) and (1),

$$\upsilon(f^{-1}) \circ \upsilon(f) = \upsilon(f \circ f^{-1}) = \upsilon(\mathrm{Id}_Y) = \mathrm{Id}_{h_Y},$$
$$\upsilon(f) \circ \upsilon(f^{-1}) = \upsilon(f^{-1} \circ f) = \upsilon(\mathrm{Id}_X) = \mathrm{Id}_{h_X},$$

Therefore, $\upsilon(f)$ is an isomorphism, and $\upsilon(f^{-1}) = \upsilon(f)^{-1}$. Conversely, suppose that $\upsilon(f)$ is an isomorphism and φ is its inverse. Let $g : Y \to X$ be the morphism suc that $\upsilon(g) = \varphi$. Then

$$\upsilon(g) \circ \upsilon(f) = \upsilon(f \circ g) = \mathrm{Id}_{h_Y}, \quad \upsilon(f) \circ \upsilon(g) = \upsilon(g \circ f) = \mathrm{Id}_{h_X},$$

which leads to

$$g \circ f = \mathrm{Id}_X, \quad f \circ g = \mathrm{Id}_Y .$$

Hence f is an isomorphism. □

1.4.4 Remark. In the case where $\mathcal{C}$ is a small category, the mapping $X \to h_X$ together with the bijections

$$\mathcal{C}(Y, X) \longrightarrow \mathrm{Nat}(F_X, F_Y)$$

defines a functor from $\mathcal{C}^{\mathrm{op}}$ to $\widehat{\mathcal{C}}$. This functor is called the *Yoneda functor*.

1.5 Representable functors

1.5.1 Definition. Let $\mathcal{C}$ be a category, F be a functor from $\mathcal{C}$ to **Set**. If there exists an object X of C such that F and h_X are isomorphic, then we say that the functor F is *representable* and that X is an object that *represents* the functor F. An isomorphism from h_X to F is called a *representation* of the functor F by the object X.

Let F be a representable functor from $\mathcal{C}$ to **Set** and X be an object of $\mathcal{C}$ that represents F. Let $\psi : h_X \to F$ be an isomorphism. Then the element $\psi_X(\mathrm{Id}_X) \in F(X)$ is called the *universal object* of the representation ψ.

1.5.2 Proposition. *Let $\mathcal{C}$ be a category, F be a functor from $\mathcal{C}$ to* **Set**. *Assume that F is representable and that $\varphi : h_X \to F$ is an isomorphism of functors that represents F, where X is an object of $\mathcal{C}$. For any object Y of $\mathcal{C}$ and any $y \in F(Y)$, there exists a unique morphism $f : X \to Y$ such that $F(f)(x) = y$. Acutally the morphism f is given by $\varphi_Y^{-1}(y)$.*

Proof. We first prove the uniqueness of f in assuming the existence. Since φ is a natural transformation from h_X to F, the diagram

$$\begin{tikzcd} h_X(X) \arrow[r, "\varphi_X"] \arrow[d, "h_X(f)"'] & F(X) \arrow[d, "F(f)"] \\ h_X(X) \arrow[r, "\varphi_Y"'] & F(Y), \end{tikzcd}$$

which can also be written as

$$\begin{tikzcd} \mathcal{C}(X,X) \arrow[r, "\varphi_X"] \arrow[d, "f\circ -"'] & F(X) \arrow[d, "F(f)"] \\ \mathcal{C}(X,Y) \arrow[r, "\varphi_Y"'] & F(Y), \end{tikzcd}$$

is commutative. Therefore, one has

$$y = F(f)(x) = F(f)(\varphi_X(\mathrm{Id}_X)) = \varphi_Y(f \circ \mathrm{Id}_X) = \varphi_Y(f),$$

which leads to

$$f = \varphi_Y^{-1}(y).$$

The same computation shows that $f := \varphi_Y^{-1}(y)$ satisfies the equality $F(f)(x) = y$. Thus the statement is proved. □

1.5.3 Corollary. *Let $\mathcal{C}$ be a category and F be a functor from $\mathcal{C}$ to* **Set**. *Then the object representing F is unique up to a unique isomorphism. Namely, if X and Y are objects of $\mathcal{C}$ representing F, and if $\varphi : h_X \to F$ and $\psi : h_Y \to F$ are corresponding representations, then there exists a unique isomorphism $f : X \to Y$ such that $F(f)$ sends the universal object of φ to that of ψ.*

Proof. Let x and y be respectively the universal objects of φ and ψ. By Proposition 1.5.2, $f := \varphi_Y^{-1}(y)$ is the unique morphism in $\mathcal{C}(X,Y)$ such that $F(f)(x) = y$. It remains to check that f is an isomorphism.

Note that, by Proposition 1.5.2, $g := \psi_X^{-1}(x)$ is the unique morphism in $\mathcal{C}(Y,X)$ such that $F(g)(y) = x$. Therefore, one has

$$F(g \circ f)(x) = (F(g) \circ F(f))(x) = F(g)(F(f)(x)) = F(g)(y) = x.$$

By Proposition 1.5.2, we obtain that $g \circ f = \mathrm{Id}_X$. Similarly, one has

$$F(f \circ g)(y) = F(f)(F(g)(y)) = F(f)(x) = y.$$

Hence, by Proposition 1.5.2, we obtain that $f \circ g = \mathrm{Id}_Y$. Therefore, f is an isomorphism. □

1.5.4 Example. Consider the forgetful functor F from **Mon** to **Set** that sends a monoid M to its underlying set, and sends a homomorphism of monoids $f : M \to N$ to itself (viewed as a mapping between sets). This functor is representable by the additive monoid $\mathbb{N}$ of natural numbers. In fact, for any element x of a monoid M, there exists a unique homomorphism of monoids $\mathbb{N} \to M$ that sends 1 to x. Therefore, the mapping

$$\varphi_M : h_{\mathbb{N}}(M) = \mathbf{Mon}(\mathbb{N}, M) \longrightarrow F(M) = M, \quad f \longmapsto f(1)$$

is a bijection, and the family $\varphi = (\varphi_M)_{M \in \mathrm{Obj}(\mathbf{Mon})}$ forms a representation of the forgetful functor M.

1.5.5 Example. Let K be a commutative unitary ring and I be a set. The functor F from $\mathbf{Mod}_K$ to $\mathbf{Set}$ that sends any K-module W to the set W^I is representable by the direct sum $K^{\oplus I}$. Recall that $K^{\oplus I}$ is defined as the set of $(a_i)_{i\in I} \in K^I$ such that $\{i \in I \,|\, a_i \neq 0\}$ is a finite set. For any K-module W, the mapping

$$W^I \longrightarrow \mathbf{Mod}_K(K^{\oplus I}, W)$$

sending $(x_i)_{i\in I}$ to

$$\big((a_i)_{i\in I} \in K^{\oplus I}\big) \longmapsto \sum_{i\in I} a_i x_i$$

is a bijection. These mappings define an isomorphism from the functor F to $h_{K^{\oplus I}}$.

1.5.6 Example. Let K be a unitary ring and V be a right K-module. Let V' be a right sub-K-module. Let F be the functor from $\mathbf{Mod}_K$ to $\mathbf{Set}$ that sends a right K-module W to the set of $f \in \mathbf{Mod}_K(V, W)$ such that

$$\forall\, x \in V',\ f(x) = 0_W.$$

Then the functor is representable by V/V'. In fact, for any $f \in F(W)$, there exists a unique homomorphism of right K-modules $\widetilde{f} : V/V' \to W$ that

$$\forall\, x \in V, \quad \widetilde{f}([x]) = f(x).$$